Claw-System: An Agentic System for DeFi Risk Monitoring and Decision Support via a Graph Time-Series Foundation Model

Anonymous ACL submission

Abstract

General-purpose LLM agents asked to recommend financial-risk interventions from raw on-chain data tend to hallucinate severity and over-trigger high-stakes actions. We design and characterise a *deployment-ready* LLM-agent pipeline for decentralised-finance supervision in which the LLM reasons not over raw transactions but over *structured forecasted evidence*: a graph time-series forecaster produces predicted exposure networks, deterministic monitors and stress scenarios convert them into a typed evidence bundle, and the LLM emits supervisory tickets with rationales. Two safety gates—a data-health gate and a confidence gate—determine when intervention-level tickets are allowed; every emitted ticket carries the evidence bundle and rationale that produced it. We release an evaluation harness that scores such pipelines on six axes, anchored to a *regulator-aligned absolute-loss* ground truth that replaces the fractional-change metric used in prior systemic-risk evaluations, and additionally measures LLM-as-judge explanation quality and a false-intervention rate. With eight reference implementations and five years of weekly DeFi exposure graphs, the full LLM-agent stack raises ticket F1 from 0.0076 to 0.0233 ($+207\%$) at a visible false-intervention cost (FIR ≈ 0.44 vs 0 for rules). A tier-above Opus 4.8 judge with a cross-family panel (Gemini 2.5 Pro, GPT-5.5) confirms the headline ordering; swapping the decision LLM for a tier-below Sonnet 4.6 *strictly* dominates Opus 4.7 on every LLM-bearing axis—F1 rises further to 0.0288, FIR collapses to 0.000, judge quality reaches 2.65, at $\sim 5\times$ lower inference cost. We characterise the resulting system as a recall–explanation Pareto option for supervisory settings, not a uniform replacement for conservative rules, and release ticket-level audit logs from the historical backtest.

1 Introduction

In May 2022 the Terra/Luna collapse erased over \$40 billion in 48 hours, triggering cascading liquidations through lending protocols, DEXs, and yield aggregators that held UST-denominated assets (Vidal-Tomás et al., 2023). Five months later FTX wiped out another \$8 billion. These crises were not isolated: in decentralised finance (DeFi), credit exposures form a dense, time-varying network of tokenised claims that no single participant fully observes, and shocks travel through it within hours.

A natural response is to build an *agent*: an LLM-driven pipeline that ingests on-chain measurements, surfaces early-warning signals, and recommends risk-mitigating actions with auditable rationale. The temptation is to point a general-purpose LLM agent at raw on-chain data and let it reason. We have found, and our reference implementations confirm (§5), that this fails in a specific and recurring way: the LLM *hallucinates severity*. It recommends high-stakes interventions even when the underlying evidence is weak, missing, or stale—grounding score 1.000, false-intervention rate 0.448. In a supervisory setting, this failure mode is far more costly than missed detections, because every intervention is observable and politically expensive.

To measure such a failure mode honestly we first need a ground truth that matches how supervisors actually prioritise. Prior systemic-risk evaluations rank protocols by *fractional* weight change (Bertomeu et al., 2024; Gonon et al., 2025; Li et al., 2025), which over-weights tiny protocols of low systemic relevance and is not what a supervisor ranks by. **We define a regulator-aligned *absolute-loss* ground truth** (§4) and use it to score, for the first time, an integrated forecast+monitor+scenario+LLM+gate pipeline against what a supervisor would actually prioritise.

The characterisation produces the first joint measurement of forecast quality, monitor lead time, and ticket quality on the same regulator-aligned target, and the first quantification of the false-intervention cost for an LLM agent operating in this setting—a cost that finance-specific LLM-agent work (Yao et al., 2026; Mao et al., 2025; Luo et al., 2025; Jeon and Lee, 2026) has not measured because none of those systems is scored against a regulator-aligned ground truth in the first place.

We close this gap with three contributions:

- **A deployment-ready LLM-agent supervisory pipeline** (Claw-System; §3). A graph time-series foundation model (Forecast-FM) produces predicted exposure networks; five deterministic monitors and five stress scenarios convert them into a typed evidence bundle; an LLM emits supervisory tickets with rationale; and two safety gates (data-health, confidence) determine when intervention-level tickets are allowed. Every emitted ticket carries the evidence bundle that produced it.

- **An evaluation harness** (Decision-Bench; §4). Six axes spanning forecast quality, streaming anomaly detection, predictive uncertainty, what-if scenario fidelity, supervisory ticket quality, and data-quality robustness, with a regulator-aligned *absolute-loss* ground truth (§4) and eight reference method implementations including persistence, EvolveGCN (Pareja et al., 2020), rule engines, snapshot LLMs, and Forecast-FM-coupled LLM variants.

- **An operating-point characterisation** (§5). On the frozen 2025 split of Exposure-Corpus, the full Forecast-FM+LLM+gate stack raises ticket F1 from 0.0076 to 0.0233 ($+207\%$). A tier-above Claude Opus 4.8 judge (Zheng et al., 2023) and a cross-family panel (Gemini 2.5 Pro, GPT-5.5) preserve the ordering $\texttt{m7}\geq\texttt{m6}>\texttt{m2}$; under the Opus 4.7 decision LLM the false-intervention rate rises to ≈ 0.44. Replacing the decision LLM with a tier-below Sonnet 4.6 *strictly dominates* Opus 4.7 on both `m6` and `m7`: F1 rises further to 0.0288 on `m7`, FIR collapses to 0.000, and judge quality reaches 2.65, at $\sim 5\times$ lower inference cost—the deployable headline operating point.

Honest scope and track fit. We have run the pipeline as a historical backtest over 283 weekly snapshots of real DefiLlama-sourced exposure data (2020–2025), not as a live production cron at a supervisory authority. The contribution is therefore best read against two of the EMNLP Industry Track's explicit acceptable categories—*methods for deployed systems (evaluation, ethics, robustness)* and *insights based on real-world datasets*—rather than as a deployment case study. The ticket-level audit logs we release come from the backtest and are intended for operators who want to recompute the operating point on their own data conditions before adopting the system.

2 Related Work

Bench positioning and ground-truth definition. LLM-agent benchmarks (HELM (Liang et al., 2022), SWE-bench (Jimenez et al., 2024), AgentBench (Liu et al., 2024)) score open-ended reasoning, software repair, and generic agent behaviour; temporal-graph benchmarks (TGB (Huang et al., 2023), OGB (Hu et al., 2020)) score structural prediction quality. Neither scores whether an LLM agent's supervisory decisions match what a regulator would actually prioritise. The closest prior systemic-risk evaluations (Bertomeu et al., 2024; Gonon et al., 2025; Li et al., 2025) rank protocols by *fractional* weight change, which disproportionately surfaces tiny protocols of low systemic relevance. Decision-Bench replaces this with an *absolute*-loss ground truth (§4) so that every downstream axis is scored against what supervisors actually rank by.

LLM agents in finance and DeFi. General agent patterns such as ReAct (Yao et al., 2023) combine reasoning with tool use; FinGPT (Yang et al., 2023) adapts language models to financial data. Within DeFi, agentic systems have so far targeted transaction auditing (Yao et al., 2026), intent mining (Mao et al., 2025), smart-contract verification (Hu et al., 2026; Kong et al., 2026), price manipulation (Liu et al., 2025), and explanation for graph anomalies (Watson et al., 2025); portfolio-oriented agents combine graph encoders with LLMs (Luo et al., 2025; Jeon and Lee, 2026). These systems consume raw transactions or token text, not structured forecasted evidence, and none reports a false-intervention rate against a regulator-aligned ground truth—a gap our characterisation (§5) closes.

Forecast-grounded LLM agents in other domains. Pairing a domain forecaster with an LLM decision layer is an emerging deployment pattern. In macroeconomics, ChatGPT-augmented

PMI nowcasting (de Bondt and Sun, 2025) and LLM-driven macroeconomic forecasting (Carriero et al., 2025) both feed structured numeric inputs into an LLM that produces interpretable narratives; a BIS primer surveys the wider pattern (Kwon et al., 2024). Time-series foundation models such as Chronos (Ansari et al., 2024), Lag-Llama (Rasul et al., 2024), and TimesFM (Das et al., 2024) make the forecaster reusable across tasks; tabular foundation models extend the same idea to heterogeneous structured data (Hollmann et al., 2025; Eremeev et al., 2025). None of these forecast→LLM pipelines, to our knowledge, has been scored against a regulator-aligned ground truth for high-stakes financial-network supervision.

3 Pipeline

Claw-System is a four-layer pipeline (Figure 1). Layer 1 forecasts the near-future exposure graph with Forecast-FM; Layer 2 converts the forecast into a typed evidence bundle of monitor alerts and stress scenarios; Layer 3 hands the bundle to an LLM that emits supervisory tickets with rationale; Layer 4 gates intervention-level actions on a data-health and a confidence check. The pipeline runs at weekly cadence on a single RTX 4090 with approximately \$10 of LLM API spend per run.

3.1 Forecaster interface

At each decision epoch t the agent observes a credit-exposure snapshot $G_t = (V, E_t, W_t)$ (an inter-protocol token-mediated exposure graph; see §A) and node covariates X_t. It queries Forecast-FM for a forecast distribution $\mathcal{P}_{t,h} \leftarrow$ Forecast-FM.Forecast(G_t, X_t, h) at horizons $\mathcal{H} = \{1, 4, 8, 12\}$ weeks, constructs an expected-weight predicted graph $\widehat{G}_{t+h}$ by thresholding edge-existence probabilities at $\pi_{\min} = 0.2$, and draws $M = 50$ Monte Carlo samples for uncertainty. Full predicted-graph construction equations are in §A.

3.2 Structured evidence

Layer 2 turns $\widehat{G}_{t+h}$ and the MC samples into a typed evidence bundle. Five *monitor functionals* score concentration, fragility, and instability of the predicted graph: top-1 PageRank (Brin and Page, 1998), HHI concentration (Rhoades, 1993), network density, PageRank Gini, and degree Gini. An alert fires when a monitor metric departs from its $L = 26$ week rolling baseline by more than $z = 2$ standard deviations; each alert is attached to (i) an attribution top-K of contributing edges by marginal contribution, and (ii) an uncertainty-aware confidence score $C_t(a) \in [0, 1]$ that decays with low data-health, wide MC dispersion, and longer horizons.

In parallel, a *scenario engine* applies five standardised stress shocks—single protocol failure, bridge cluster failure, stablecoin de-peg, sector lending shock, correlated top-10 stress—to both $\widehat{G}_{t+h}$ and each MC sample, and aggregates the worst-half losses as $\mathrm{CVaR}_{0.5}(s, h)$. Full monitor and scenario definitions are in §B.

3.3 LLM decision layer

Layer 3 receives the evidence bundle as a typed JSON payload—alerts with attribution top-K, scenario CVaR table, MC dispersion, and the scalar data-health score DH_t—and is asked to emit a ranked list of supervisory tickets. Each ticket carries (a) a severity level drawn from a fixed four-action playbook (`Monitor`, `Investigate`, `Recommend-Reduce`, and `Contingency`), (b) a target set of protocols, and (c) a free-form rationale that cites specific evidence fields by name. We additionally probe self-consistency by repeating the call three times at temperature 0 and reporting Jaccard overlap on targets and severities.

3.4 Safety gates

Two gates determine the feasibility of intervention-level actions. The *data-health gate* $DH_t = \mathrm{mean}(\mathrm{fresh}_t, \mathrm{miss}_t, \mathrm{topo}_t, \mathrm{disc}_t) < \tau_{\mathrm{data}} = 0.7$ triggers safe mode, in which only `Monitor` and `Investigate` tickets are emitted, regardless of LLM recommendation. The *confidence gate* $\bar{C}_t < \tau_{\mathrm{conf}} = 0.6$ blocks intervention-level tickets even when data is healthy. Every emitted ticket carries the full evidence bundle, the LLM rationale, and the gate states that produced it; the audit log is the unit of release (§6).

The end-to-end algorithm is in §B (Algorithm 1).

4 Evaluation Harness

Why a new harness. Scoring an LLM-agent supervisory pipeline requires evidence on six independent axes that no existing benchmark covers together: how good is the forecast (`b1_forecast`), how early does the monitor turn (`b2_warning`), is the predictive uncertainty calibrated (`b3_calibration`), do scenario losses match reality (`b4_stress`), do the emitted tickets match what a regulator would prioritise

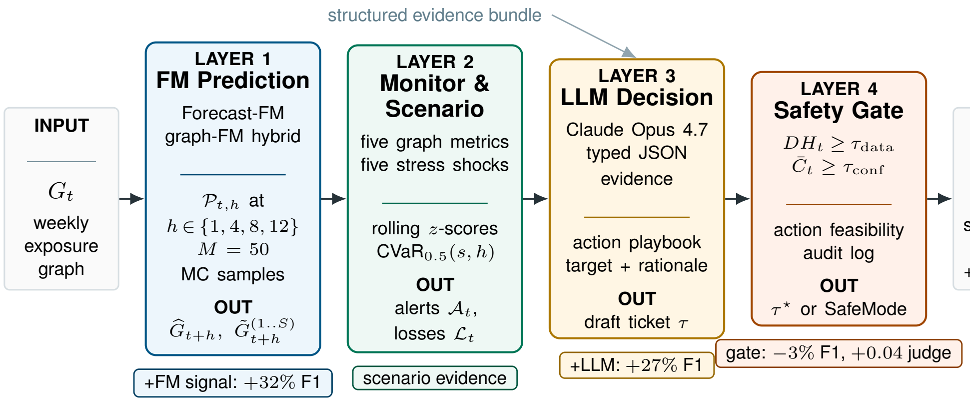

Figure 1: Claw-System: a four-layer pipeline. The agent ingests the weekly exposure graph G_t, runs FM prediction (Layer 1), monitor and scenario engines (Layer 2), an LLM decision layer (Layer 3), and a safety gate (Layer 4) that emits either the supervisory ticket $\tau^\star$ or a safe-mode notice. Bottom annotations report the per-layer F1 / judge lift from §5.

(b5_decision), and what happens under degraded data (b6_robustness). LLM-agent benchmarks (Liang et al., 2022; Jimenez et al., 2024; Liu et al., 2024) cover none of these for our setting; temporal-graph benchmarks (Huang et al., 2023; Hu et al., 2020) cover only structural prediction quality. Decision-Bench unifies the six axes on a shared dataset and reference-method set so that ablations on the forecaster, monitor, scenario, and decision layers can be reported independently.

Dataset and split. We use the Exposure-Corpus dataset of weekly directed weighted exposure graphs (4,300 protocols, 24,300 tokens, 43.7M exposure entries, 283 weekly snapshots from 2020-03 to 2025-08 (Wu et al., 2025)). We adopt a strict no-leakage temporal split: 2020-03–2024-06 train ($\sim$222 weeks), 2024-07–2024-12 validation ($\sim$26 weeks, used for conformal calibration and threshold tuning), 2025-01–2025-08 test ($\sim$33 weeks, frozen leaderboard). For b2_warning we additionally run a historical event study around Terra/Luna (2022-05-09), FTX (2022-11-07), and SVB/USDC (2023-03-10); the event study is reported separately from the 2025 predictor-specific leaderboard.

Regulator-aligned absolute-loss ground truth. Prior systemic-risk evaluations rank protocols by *fractional* weight change, which silently upweights tiny protocols of low systemic relevance. Decision-Bench ranks instead by *absolute* loss: with $w_t(v)$ the total weight incident on v at t, the stressed set at horizon h is $\mathcal{S}_t^h = \mathrm{top}_\pi\{v : w_t(v) > 0,\ \Delta_t^h(v) > 0\}$, where $\Delta_t^h(v) = w_t(v) - w_{t+h}(v)$. We fix $\pi = 0.05$ and $W = 4$ weeks throughout (yielding $|\mathcal{S}_t^h| \approx 283$ protocols per week, stable across the test split). The same ground truth anchors b2_warning, b5_decision, and the LLM-decision evaluation pipeline.

Eight reference implementations. The harness bundles eight reference methods spanning the four ablation cuts of the four-layer framework: heuristic streaming monitor (h1), persistence+rules (m1), snapshot LLM (m2), EvolveGCN (Pareja et al., 2020) (m3), Forecast-FM-only (m4), Forecast-FM+rules (m5), Forecast-FM+LLM (m6), and Forecast-FM+LLM+gate (m7). Each method exposes the same `predict_graph` or `decide_actions` interface so adding a ninth method is a single-file change. The full method specification and per-axis schema are in §C–§D.

Reproducibility. A complete leaderboard run finishes in under four GPU-hours on a single RTX 4090 plus approximately \$10 of LLM API spend, making the harness accessible without industrial compute. The harness fixes random seeds, snapshots library versions, and emits SHA-256 hashes of all checkpoint files; every reported number can be reproduced bit-exactly from the released artefacts.

5 Empirical Results

We use Decision-Bench to ask three questions about the pipeline. First, does Forecast-FM provide forecast signal that persistence does not (b1, b3, b6)? Second, does routing the LLM through structured evidence improve supervisory tickets once precision, recall, explanation quality, and false interventions are measured jointly (b4, b5)? Third, which components are load-bearing and which are reserves under degradation (ablations)? All headline numbers come from the frozen 2025 test split.

Forecast-FM contributes trend, calibration, and robustness; persistence still wins on static point error. On static error metrics, persistence is a deceptively strong baseline: at $h = 4$ it matches the FM on PageRank MAE (3.4×10^{-5} vs 4.5×10^{-5}) and Spearman rank correlation (0.570 vs 0.558). The FM contributes signal where persistence is structurally pinned: trend consistency 0.525–0.632 across horizons (persistence: 0); `b3_calibration` PI coverage 0.913 against a 0.90 target and ECE 0.013; `b6_robustness` mean relative degradation 0.113 vs 0.148 for persistence (24% less degradation under data corruption). The full per-horizon and per-regime audits are in §E.

Headline decision-quality result. Table 1 consolidates the central `b5_decision` comparison. The persistence+rules baseline (`m1`) achieves the highest single-axis precision (0.720, in §E) and a false-intervention rate of 0 by virtue of raising few tickets. Against this baseline, the full Forecast-FM+LLM+gate stack (`m7`) lifts ticket F1 from 0.0076 to 0.0233 (+208%). Against the comparable pure-LLM baseline that consumes raw snapshots without the forecaster (`m2`, the only other LLM-bearing row), `m7` lifts LLM-as-judge explanation quality from 2.24 to 2.45/5 (+9.4%) under a Claude Opus 4.8 judge that is tier-above the Claude Opus 4.7 decision LLM, satisfying the standard judge-stronger constraint of the LLM-as-judge literature (Zheng et al., 2023). The cost surfaces sharply once `m2` is in the table: adding the FM signal to the LLM (`m2`↠`m6`) raises the false-intervention rate from 0.000 to 0.448. The safety gate (`m6`↠`m7`) reduces FIR only marginally ($0.448 \to 0.437$); under the stronger judge the gate's effect on judge score is also marginal (+0.04), revealing that the +0.21 gain reported under the weaker Claude Haiku 4.5 judge in prior pilots was partially a ceiling artifact. Both LLM variants reach grounding 1.000, so the failure mode is not citation hallucination but *over-reading the forecaster's evidence as warrant for high-severity action.*[1]

[1] `m3_evolvegcn` and `m4_fm_only` are predictor-only and do not emit tickets; `m3`'s `b1_forecast` numbers are in §E.1. EvolveGCN underperforms persistence on the downstream metrics in our regime, consistent with the data efficiency limit (∼220 weekly training snapshots).

Judge robustness across model families. We re-judge `m2`/`m6`/`m7` with a cross-family panel (Table 2). All three frontier judges preserve the headline ordering `m7`≥`m6`>`m2` on the released configuration; `m7`'s mean score is within 0.45 across all three. The largest disagreement is that Gemini 2.5 Pro scores `m2` slightly above `m6` (2.69 vs 2.55), still keeping `m7` on top. The cross-family agreement closes the judge-family confound a single Anthropic judge would leave open.

A tier-below, cheaper decision LLM dominates the headline operating point. We re-run the decision layer of `m6` and `m7` on the *same* FM-derived prompts with two non-Opus 4.7 decision LLMs—Claude Sonnet 4.6 (within-family, tier-below, ∼5× cheaper) and Google Gemini 2.5 Pro (cross-family)—and re-score with the same Opus 4.8 judge (Table 3). On both architectures, Sonnet 4.6 is a strict Pareto improvement over Opus 4.7: F1 rises (+15% on `m6`, +24% on `m7`), the false-intervention rate collapses from 0.437–0.448 to 0.000, and judge quality improves (+0.11 on `m6`, +0.20 on `m7`). The cross-family Gemini 2.5 Pro decision is more conservative still (lower F1, FIR = 0.000), confirming that the over-intervention failure mode is specific to Opus 4.7 rather than a generic LLM-decision artefact. The headline operating point a practitioner would now deploy is `m7` with Sonnet 4.6: F1 = 0.0288, FIR = 0.000, and judge = 2.65—higher than every row in Table 1 on every axis at a fraction of the inference cost.

Layer-wise contribution. Holding the decision policy fixed at *LLM* and adding the Forecast-FM signal lifts ticket F1 by 32% ($0.0183 \to 0.0241$); holding the predictor fixed at Forecast-FM and swapping rules for the LLM lifts F1 by another 27% ($0.0189 \to 0.0241$). The two layers therefore make additive, non-substitutable contributions (Figure 2). Adding the safety gate trades 3% F1 for a small judge-quality change (+0.04 under Opus 4.8, +0.35 under Gemini 2.5 Pro, 0 under GPT-5.5; Table 2); under the stronger judge panel the gate's primary value is operational (every emitted intervention-level ticket clears two pre-defined gates with auditable logs) rather than expressed as a single explanation-quality delta.

Ablation: load-bearing vs reserve components. Under clean 2025 data the confidence gate is load-bearing: removing it raises FIR from 0.000 to 0.429. The scenario engine is load-bearing for

ID	Pred.	Decision	F1↑	Recall↑	Judge↑	FIR↓	Ground.↑	Stabil.↑
m1	persist.	rules	0.0076	0.004	—	**0.000**	—	0.514
m2	—	LLM (no FM)	0.0183	0.009	2.24	**0.000**	1.000	0.532
m5	Forecast-FM	rules	0.0190	0.010	—	**0.000**	—	0.257
m6	Forecast-FM	LLM	**0.0241**	0.012	2.41	0.448	1.000	0.488
m7	Forecast-FM	LLM + safety gate	0.0233	0.012	**2.45**	0.437	1.000	0.435

Table 1: Headline `b5_decision` comparison on the frozen 2025 test split. Rows are read as a multi-objective trade-off rather than a single leaderboard: `m1` is the conservative high-precision baseline; `m2` is the pure-LLM baseline (no forecaster); `m6`/`m7` are the FM-grounded LLM variants. Judge scores are from a Claude Opus 4.8 judge (within-family upgrade, strictly tier-above the Claude Opus 4.7 decision LLM). Cross-family judge agreement is in Table 2. `m3_evolvegcn` and `m4_fm_only` are predictor-only and do not emit tickets; their forecast quality is reported in §E.1.

Method	Opus 4.8	Gemini 2.5 Pro	GPT-5.5	Haiku 4.5[†]
m2	2.24	2.69	1.90	2.03
m6	2.41	2.55	2.45	2.48
m7	**2.45**	**2.90**	2.45	2.69

Table 2: Cross-family judge panel on the same Opus 4.7 decisions ($n = 29$). Gemini and GPT use $\text{reasoning} = \{\text{effort} : \text{minimal}\}$ so the thinking budget matches the Anthropic baseline. [†] Haiku 4.5 is a tier-below weak-judge reference.

Method	Decision LLM	F1↑	FIR↓	Judge↑
m6	Opus 4.7 (main)	0.0241	0.448	2.41
m6	Sonnet 4.6	**0.0276**	**0.000**	**2.52**
m6	Gemini 2.5 Pro	0.0129	0.000	2.21
m7	Opus 4.7 (main)	0.0233	0.437	2.45
m7	Sonnet 4.6	**0.0288**	**0.000**	**2.65**
m7	Gemini 2.5 Pro	0.0139	0.000	2.38

Table 3: Cross-decision-LLM panel on the same FM-derived prompts, judge fixed at Claude Opus 4.8 ($n = 29$). Sonnet 4.6 strictly dominates Opus 4.7 on both architectures; Gemini 2.5 Pro is more conservative still. The bold row, `m7` × Sonnet 4.6, outperforms every row in Table 1 on every LLM-bearing axis at ∼5× lower inference cost.

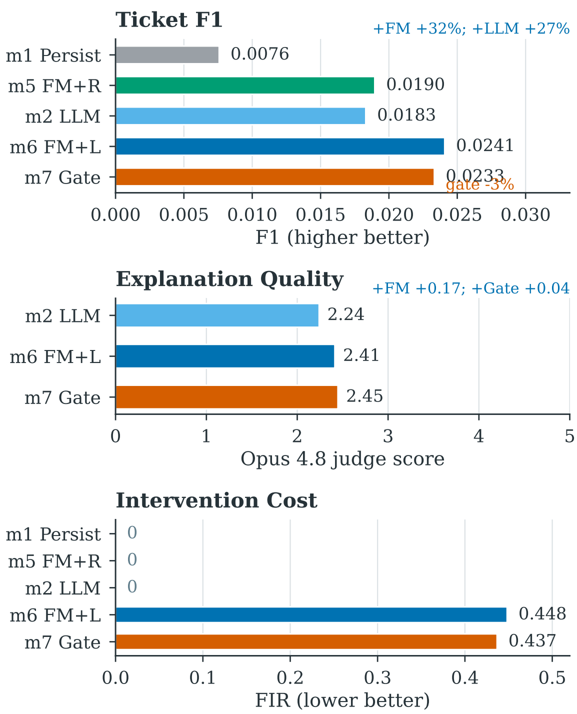

Figure 2: Layer-wise contribution and cost. Adding the FM signal to a snapshot LLM improves ticket F1 and explanation quality; swapping the FM-fed rule engine for the LLM improves recall-oriented F1 further; the same LLM variants incur a substantial false-intervention rate, partially mitigated by the safety gate.

target extraction: skipping it collapses ticket precision from 0.600 to 0.000. The data-health gate and multi-horizon diversity are dormant on clean data but activate as reserves under degradation. Under severe feature/edge masking (80–98%) the data-health gate suppresses up to 60% of interventions that would otherwise produce false alarms (FIR range $[0.27, 0.60]$ when disabled vs 0.000 when triggered); multi-horizon monitoring generates ∼4× more alerts in crisis windows (Terra/Luna, FTX, SVB/USDC) at unchanged precision. Per-regime ablation audits are in §E.6.

Early-warning timeline (historical event study). For the three pre-test crisis windows the weighted-degree monitor `h1_weighted_degree` achieves median lead times of 4–5 weeks with precision 1.000 at SVB/USDC across all alert budgets $K \in \{5, 10, 20\}$; per-budget details in §E.4. This is historical monitor evidence, not an FM-specific test-split result.

6 Deployment Lessons

We surface four lessons from building Claw-System that we expect to transfer to other forecast→LLM pipelines for high-stakes structured-evidence decisions.

L1. Grounding $\neq$ correctness. Our LLM variants achieve grounding score 1.000: every cited number in every emitted ticket is traceable to a field of the input evidence bundle. They still pay $\mathrm{FIR} = 0.448$. The failure is not that the LLM invents evidence; it is that the LLM *over-reads* weak-but-present evidence as a justification for high-severity action. Grounding tells you the agent did not hallucinate citations; only a downstream metric that measures intervention realism (FIR against an absolute-loss ground truth) tells you whether the agent is safe to act on.

L2. Audit logs are a design artefact, not an output. Every emitted ticket carries the forecast distribution, monitor scores, scenario CVaRs, the LLM rationale, the gate states, and the seeds that produced it. We release these ticket-level audit logs because the operating point we report is not what every operator will need: a more conservative supervisor will retune τ_{data} or τ_{conf} and rerun the decision policy without re-invoking the FM or the LLM. Audit logs treated as a first-class artefact make the system retunable post-hoc.

L3. Safety gates are reserves, not always-on filters. On clean 2025 data the data-health gate is dormant (§5); its impact is invisible to a leaderboard that does not include a degradation track. Under masking regimes, the gate suppresses up to 60% of FIR-eligible tickets at the cost of suspending intervention-level actions. The deployable lesson is to evaluate gates on dedicated stress regimes rather than on clean test splits; if you only evaluate on the latter you will conclude the gate is unnecessary and ship without it.

Limitations

Backtest, not field deployment. All results come from a historical backtest over 283 weekly snapshots of Exposure-Corpus (2020–2025). We have not run Claw-System as a live weekly cron at a supervisory authority. The ticket-level audit logs we release come from the backtest. Field deployment lessons—operator behaviour, end-to-end latency under real data feeds, regulator-in-the-loop calibration—are out of scope for this paper and remain the most pressing future direction.

Single domain. The bench evaluates a single on-chain risk surface (DeFi inter-protocol credit exposure). Generalisation to NFT lending, perpetual derivatives, or cross-chain bridge networks is untested and will require domain-specific re-calibration of the conformal split and the stressed-set percentile π.

Weekly resolution. The pipeline operates at weekly granularity, but several DeFi shocks unfold within hours—Terra/Luna erased $40B in under 48 hours. A sub-hourly production deployment would require denser snapshot ingestion and re-evaluation of conformal calibration on the resulting denser time series. The 4–5 week lead times we report should be read as "how early the weekly monitor turns", not as end-to-end alert latency.

Judge thinking-budget is held minimal. The cross-family judge panel (Claude Opus 4.8, Google Gemini 2.5 Pro, OpenAI GPT-5.5) is run with reasoning effort set to *minimal* so the comparison against the non-reasoning Anthropic baseline is apples-to-apples on intrinsic model capability rather than thinking budget. Default reasoning mode on Gemini 2.5 Pro and GPT-5.5 would change the reported scores; we treat the reasoning-on configuration as a distinct evaluation surface and leave it for a follow-up study. Practitioners deploying the judge at default reasoning settings should re-calibrate the score thresholds reported here. Human-judge calibration with practising supervisors remains the most pressing methodological extension.

Offline decision evaluation. `b5_decision` scores tickets against a regulator-aligned absolute-loss ground truth, which captures supervisory targeting priority but not the downstream effect of supervisory action—whether the recommended risk reduction would actually have averted contagion. A controlled live evaluation with a regulator in the loop, which the bench approximates but cannot replace, remains the primary direction we plan to extend the harness in.

Ethics Statement

Data provenance. Exposure-Corpus is built from public on-chain measurements aggregated via DefiLlama (DefiLlama, 2026). No private user data, wallet-level personally-identifying information, or off-chain identity data is used. Protocol-level aggregates are the unit of analysis throughout.

Decision risk. Claw-System emits supervisory tickets, not automated trades or automated interventions. The high-severity action types (`Recommend-Reduce`, `Contingency`) are gated on a data-health score and a confidence score, and the pipeline emits a safe-mode notice when either gate fails. We characterise the system as decision support for a human supervisor, not as an autonomous risk agent. The measured false-intervention rate ($\mathrm{FIR} \approx 0.44$ for the Forecast-FM+LLM variant) is reported in the headline results table so that downstream operators can weigh the trade-off against their own risk tolerance before deployment.

LLM use disclosure. The decision-layer LLM (`m2`, `m6`, `m7`) is Claude Opus 4.7 in the main configuration; Claude Sonnet 4.6 and Google Gemini 2.5 Pro are reported as decision-layer ablations. The LLM-as-judge for the explanation-quality axis is a three-model cross-family panel: Claude Opus 4.8 (within-family upgrade, strictly tier-above Opus 4.7), Google Gemini 2.5 Pro, and OpenAI GPT-5.5 (cross-family). All judges are run with reasoning effort set to *minimal* so the comparison is controlled for thinking budget. Claude Haiku 4.5 from a prior run is reported in the appendix as a weak-judge baseline. Every model is accessed through the OpenRouter unified API at temperature 0.

Release. We will release the harness, the eight reference implementations, and the ticket-level audit logs under a permissive open licence. Model checkpoints will be released subject to dataset-licence compatibility checks.

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A Forecaster Interface and Notation

A.1 Credit exposure

Let $X_G(q)$ be the set of tokens issued by protocol q and $X_p(t)$ the set of tokens held by protocol p at time t. Protocol p has a credit exposure to protocol q at t if $X_p(t) \cap X_G(q) \neq \emptyset$ (Wu et al., 2025). This token-mediated dependence generalises counterparty credit risk from traditional finance to DeFi.

A.2 Network representation

Over an interval $\tau = t_2 - t_1$, exposures form a weighted directed graph $G_\tau = (P_\tau, E_\tau)$. Node weights are $w_\tau(p) = \sum_{\sigma \in X_p(t_1) \cap X_p(t_2)} v_{t_2,\sigma}$ where $v_{t_2,\sigma}$ is the USD value of token σ at t_2; edge weights aggregate non-negative token flows from p to q mapped to issuer q, as defined in Wu et al. (2025). We adopt the same construction without modification.

A.3 Forecast primitive and predicted graph

At each decision epoch t the agent queries $\mathcal{P}_{t,h} \leftarrow \text{Forecast-FM.Forecast}(G_t, X_t, h)$. From $\mathcal{P}_{t,h}$ we build a representative predicted graph via a thresholded expected-weight operator,

$$\widehat{w}_{pq,t+h} := \Pr(e_{pq,t+h} = 1) \cdot \mathbb{E}[w_{pq,t+h} \mid e_{pq,t+h} = 1], \tag{1}$$

$$\widehat{E}_{t+h} := \{(p,q) : \Pr(e_{pq,t+h} = 1) > \pi_{\min}\}, \tag{2}$$

with $\pi_{\min} = 0.2$ (tuned on validation). For uncertainty quantification we draw $M = 50$ Monte Carlo samples $\{\widetilde{G}^{(m)}_{t+h}\}$ by Bernoulli edge sampling and Gaussian weight perturbation of $\mathcal{P}_{t,h}$.

A.4 Hybrid prediction strategy

In practice Forecast-FM keeps all edges from G_t, reweights them as $w'_{pq} = w_{pq} + \Pr(e_{pq,t+h} = 1) \cdot \widehat{r}_{pq}$ where $\widehat{r}_{pq}$ is the FM residual, and additionally adds new edges whose existence probability exceeds $\pi_{\min}$. This preserves network structure while letting the FM adjust weights and predict new connections.

A.5 Architecture and training

Forecast-FM is built on the GraphPFN graph-tabular foundation backbone (Hollmann et al., 2025; Eremeev et al., 2025) with a LiMiX-16M tabular transformer encoder: 12 layers, 4 attention heads, hidden dimension 64. The same backbone serves three heads—edge existence (binary), edge weight (regression), and node-TVL change (regression)—trained jointly with a BCE+MSE composite loss. We fine-tune one checkpoint per horizon $h \in \{1, 4, 8, 12\}$ for 20 epochs with AdamW (learning rate 10^{-4} for heads, 10^{-5} for the backbone). Training uses 104 weekly snapshots (March 2020–January 2024), validation uses 12 weeks (February–April 2024), and the leaderboard test split is the remaining 33 weeks (§C). Full-graph in-context inference fits a single RTX 4090; one weekly forecast across all four horizons completes in well under a minute.

Held-out link-prediction quality on the validation split is AUPRC $= 0.978$ / 0.973 / 0.967 / 0.967 and AUROC $= 0.996$ / 0.995 / 0.993 / 0.993 at $h = 1$ / 4 / 8 / 12 respectively; a frozen GraphPFN baseline (no fine-tuning) achieves AUPRC 0.936–0.940 at the same horizons, isolating the contribution of domain fine-tuning. These intrinsic forecaster numbers do not appear in the main text because the paper's contribution is the operating point of the full pipeline, not the FM's headline accuracy.

B Pipeline Mathematics

B.1 Data-health gate

The data-health score combines four deterministic checks of the incoming snapshot:

$$DH_t = \text{mean}\big(\text{fresh}_t,\ \text{miss}_t,\ \text{topo}_t,\ \text{disc}_t\big) \in [0, 1]. \tag{3}$$

Safe mode is triggered when $DH_t < \tau_{\text{data}} = 0.7$: monitor alerts are still emitted but intervention-type tickets are suppressed.

B.2 Monitor metrics

ID	Metric	Definition
N1	Systemic Importance	$\max_{p\in V}$ PageRank$(\widehat{G}_{t+h})_p$
N2	HHI Concentration	$\sum_p (d_p/\sum_q d_q)^2$, d_p weighted out-degree
N3	Network Density	$\lvert E_{t+h}\rvert/\lvert V\rvert(\lvert V\rvert-1)$
N4	PageRank Gini	Gini$(\{\mathrm{PR}_p\}_{p\in V})$
N5	Degree Gini	Gini$(\{d_p\}_{p\in V})$

PageRank uses power iteration with damping 0.85 and 10 iterations. An alert fires when $|N_{t,h}^{(j)} - \mu_t^{(j)}|/\max(\sigma_t^{(j)}, \epsilon) > z$ with $z = 2$, $L = 26$ week rolling baseline, $\epsilon = 10^{-8}$.

B.3 Uncertainty-aware alert confidence

With $\mathrm{CV}_{t,h}^{(j)}$ the coefficient of variation of metric j across the M MC samples (clamped to $[0, 1]$),

$$C_t(a) := DH_t \cdot \frac{1}{1 + \mathrm{CV}_{t,h}^{(j)}} \cdot \frac{1}{1 + \ln(1 + h)} \in [0, 1]. \tag{4}$$

$C_t(a)$ decreases monotonically with lower data health, wider predictive spread, and longer horizons.

B.4 Scenario engine

ID	Scenario	Target	Shock
S1	Single protocol fail	Highest-weight node	100%
S2	Bridge cluster fail	All bridge nodes	100%
S3	Stablecoin de-peg	All stablecoin nodes	50%
S4	Sector lending shock	All lending nodes	30%
S5	Correlated stress	Top-10 by weight	20%

Per-scenario loss $L_{t,h}(s) = (\sum_e w_e(\widehat{G}_{t+h}) - \sum_e w_e(\widehat{G}^s_{t+h}))/\sum_e w_e(\widehat{G}_{t+h})$; CVaR aggregation averages the worst $\lceil \lambda M \rceil$ MC-sample losses with $\lambda = 0.5$.

B.5 Decision feasibility and ticket scoring

An intervention-level action u is feasible iff

$$\big(\mathrm{SAFE\_MODE}_t = 0\big) \;\wedge\; \big(\bar{C}_t \geq \tau_{\mathrm{conf}} = 0.6\big), \tag{5}$$

with $\bar{C}_t$ the mean alert confidence. Feasible tickets are scored as $\mathrm{Score}_t(u) = w_u \cdot \bar{C}_t \cdot \mathrm{Imp}_t$, where $w_u \in \{0.25, 0.50, 0.75, 1.00\}$ is the playbook severity weight and Imp_t is the mean CVaR across scenarios (default 1.0 when no scenarios are available).

B.6 End-to-end procedure

Algorithm 1 Claw-System: one decision epoch.

Require: (G_t, X_t); horizons $\mathcal{H}$; scenarios $\mathcal{S}_0$; thresholds $(\tau_{\mathrm{data}}, \tau_{\mathrm{conf}}, z, \pi_{\min})$; MC count M
Ensure: Alerts $\mathcal{A}_t$; scenario summary $\mathcal{R}_t$; tickets $\mathcal{D}_t$

$DH_t \leftarrow \mathrm{DataHealth}(G_t, X_t)$; $\mathrm{SAFE\_MODE}_t \leftarrow \mathbb{I}\{DH_t < \tau_{\mathrm{data}}\}$
for all $h \in \mathcal{H}$ **do**
  $\mathcal{P}_{t,h} \leftarrow$ Forecast-FM.Forecast(G_t, X_t, h)
  $\widehat{G}_{t+h} \leftarrow \mathrm{BuildPredGraph}(\mathcal{P}_{t,h}; \pi_{\min})$
  $\{\widetilde{G}^{(m)}_{t+h}\} \leftarrow \mathrm{MCSample}(\mathcal{P}_{t,h})$
  Compute $\mathbf{N}_{t,h} = \Phi(\widehat{G}_{t+h}, X_t)$, attach rolling stats, derive alerts $\mathcal{A}_{t,h}$ and confidences $C_t(\cdot)$
  for all $s \in \mathcal{S}_0$ **do**
    Compute $L_{t,h}(s)$, $\widetilde{L}^{(m)}_{t,h}(s)$, $\mathrm{CVaR}_{0.5}(s, h)$
  end for
end for
$\mathcal{D}_t \leftarrow \mathrm{PlaybookDecision}(\mathcal{A}_t, \mathcal{R}_t, DH_t, \mathrm{SAFE\_MODE}_t, \tau_{\mathrm{conf}})$
return $(\mathcal{A}_t, \mathcal{R}_t, \mathcal{D}_t)$

C Benchmark Details

C.1 Six-axis schema

ID	Capability tested	Primary metrics
`b1_forecast`	Temporal graph prediction	PageRank/HHI/Gini MAE, trend consistency, Spearman; per $h \in \{1, 4, 8, 12\}$
`b2_warning`	Streaming anomaly detection	Precision, recall, F1, lead time, alert stability; per event × budget
`b3_calibration`	Predictive uncertainty	PI coverage at 0.90 target, PI width, ECE, CRPS
`b4_stress`	What-if scenario fidelity	Loss MAE, distressed-count MAE, propagation depth MAE, target overlap@k
`b5_decision`	Supervisory ticket quality	Precision, recall@k, F1, target stability, judge score, FIR
`b6_robustness`	Data quality sensitivity	Per-benchmark metrics under 5 degradation regimes; relative degradation

The six axes are chosen to be (i) *non-overlapping*—a system can excel at `b1` while failing `b5`; (ii) *decomposable along the four-layer framework*, so that ablations on the predictor layer (`b1`, `b3`, `b6`), monitor layer (`b2`), scenario layer (`b4`), and decision layer (`b5`) can be reported independently; and (iii) *individually publishable*, so that downstream work may extend a single axis without re-running the whole suite.

C.2 Dataset split

Split	Period	Weeks	Usage
Train	2020-03–2024-06	∼222	Forecaster training
Validation	2024-07–2024-12	∼26	Conformal calibration, threshold tuning
Test	2025-01–2025-08	∼33	Frozen leaderboard

C.3 Ground-truth equation

With $w_t(v) = \sum_{e \ni v} w_t(e)$ the total weight incident on v, the regulator-aligned stressed set at horizon h is

$$\Delta_t^h(v) = w_t(v) - w_{t+h}(v), \quad \mathcal{S}_t^h = \mathrm{top}_\pi \{ v : w_t(v) > 0,\ \Delta_t^h(v) > 0 \}, \tag{6}$$

with $\pi = 0.05$ and lookahead window $W = 4$ weeks. This is the single ground-truth definition used by `b2`, `b5`, and the LLM-decision evaluation pipeline.

C.4 Reporting protocol

A submission produces one JSON file per {benchmark, method} pair following the harness schema, plus a single `results.json` aggregating the suite. The full leaderboard requires `b1`, `b3`, `b5`, `b6`; `b2` and `b4` are required only for systems that implement a monitor or scenario layer respectively. The harness fixes random seeds, snapshots library versions, and emits SHA-256 hashes of all checkpoint files.

D Reference Methods

ID	Predictor	Policy	Role
`h1_weighted_degree`	—	heuristic alert	Streaming anomaly baseline (b2)
`m1_persistence_rules`	$G_{t+h} = G_t$	rule engine	Decision baseline
`m2_snapshot_llm`	current snap	LLM	LLM-only ablation (no forecast)
`m3_evolvegcn`	EvolveGCN	—	Learned-GNN baseline
`m4_fm_only`	Forecast-FM	—	FM-only ablation
`m5_fm_rules`	Forecast-FM	rule engine	FM + rules
`m6_fm_llm`	Forecast-FM	LLM	Full FM + LLM stack
`m7_fm_llm_gated`	Forecast-FM	LLM + safety gate	Safety-gated FM + LLM

The eight methods expose all four cuts of the four-layer ablation: adding the predictor layer (`m1`↠`m5`), swapping the decision layer (`m5`↠`m6`), adding the safety gate (`m6`↠`m7`), and removing the predictor from the LLM stack (`m2`↞↠`m6`). All methods conform to a shared `predict_graph/decide_actions` interface; adding a ninth method is a single-file change.

LLM model panel and decoding. All LLM-bearing methods (`m2`, `m6`, `m7`) are evaluated under three decision-layer configurations: Claude Opus 4.7 (main), Claude Sonnet 4.6 (within-family, tier-below ablation), and Google Gemini 2.5 Pro (cross-family ablation). Each runs at temperature 0 and

$\max\_tokens = 4096$; self-consistency is probed with three repeated calls per week and Jaccard overlap on emitted ticket targets and severities.

The LLM-as-judge for the explanation-quality axis is a three-model panel chosen so the judge tier is at or above the decision tier on the public 2026 Artificial Analysis index: Claude Opus 4.8 (within-family upgrade, strictly above Opus 4.7), Google Gemini 2.5 Pro, and OpenAI GPT-5.5 (cross-family upgrades). The Gemini and GPT judges are reasoning-by-default models; we attach OpenRouter's $\mathrm{reasoning} = \{\mathrm{effort} : \mathrm{minimal}\}$ control to both so the judge thinking budget matches the non-reasoning Anthropic baseline. Claude Haiku 4.5 from a pilot run is reported as a weak-judge reference. Every model is served through the OpenRouter unified API; observed cost is on the order of a few US dollars per full sweep of the 29-week test split.

E Full Per-Slice Results

This appendix expands the headline numbers of §5 into the per-slice audits underlying them.

E.1 Full all-horizon `b1_forecast` results

Table 4: `b1_forecast` all-horizon forecasting on the 2025 test split. PageRank MAE shown in 10^{-5} units. Best per horizon in **bold**. Trend consistency for `m1_persistence` is structurally 0 because $\hat{G}_{t+h} = G_t$ produces no trend signal.

h	Method	PR MAE (↓)	HHI MAE (↓)	Gini MAE (↓)	Rank Corr (↑)	Trend Cons (↑)
1	`m1_persistence`	**3.6**	0.0376	**0.0027**	**0.556**	0.000
	`m3_evolvegcn`	3.7	0.0481	0.1149	0.535	0.444
	`m4_fm_only`	4.7	0.0376	0.0029	0.549	**0.531**
	`m5_fm_rules`	4.7	0.0376	0.0029	0.549	0.525
4	`m1_persistence`	**3.4**	0.0322	**0.0018**	**0.570**	0.000
	`m3_evolvegcn`	3.5	0.0489	0.1326	0.551	0.324
	`m4_fm_only`	4.5	**0.0321**	0.0022	0.559	0.621
	`m5_fm_rules`	4.5	**0.0321**	0.0022	0.558	**0.628**
8	`m1_persistence`	**3.9**	0.0399	**0.0028**	**0.517**	0.000
	`m3_evolvegcn`	4.0	0.0492	0.1140	0.499	0.296
	`m4_fm_only`	4.5	0.0398	0.0030	0.508	0.624
	`m5_fm_rules`	4.5	0.0398	0.0030	0.508	**0.632**
12	`m1_persistence`	**3.7**	0.0393	**0.0027**	**0.525**	0.000
	`m3_evolvegcn`	3.8	0.0534	0.1245	0.509	0.314
	`m4_fm_only`	4.1	**0.0392**	0.0029	0.516	**0.600**
	`m5_fm_rules`	4.1	**0.0392**	0.0029	0.516	**0.600**

E.2 Per-scenario `b4_stress` results

E.3 Per-regime `b6_robustness` results

E.4 Per-budget `b2_warning` results

E.5 Per-regime A1 (data-health gate) isolation

E.6 Stress-condition ablation Panels B and C

Panel B: under severe feature/edge masking, FIR ranges over $[0.27, 0.60]$ when the data-health gate is disabled ($\tau_{\mathrm{conf}} = 0$), and collapses to 0.000 when the gate is triggered, by suppressing intervention tickets. Panel C: multi-horizon monitoring ($h \in \{1, 4, 8, 12\}$) generates 73–98 alerts in the three crisis windows vs 19–23 for single-horizon ($h = 4$), at unchanged precision range 0.533–0.624 and zero false interventions in both cases.

Table 5: `b4_stress` per-scenario detail. Loss MAE in normalised units; Distress count MAE in protocols; target overlap@10 measured against the actual future graph. Best per scenario row in **bold**. The main pattern is conservative: FM variants roughly match the strong persistence baseline on loss and target-overlap metrics, while EvolveGCN is generally weaker on downstream what-if fidelity.

Scenario	Method	Loss MAE (↓)	Distress MAE (↓)	Prop. Depth MAE (↓)	Overlap@10 (↑)
S1 (single protocol)	`m1_persistence`	0.0668	**200.3**	0.897	**0.314**
	`m3_evolvegcn`	0.1072	218.1	0.966	0.034
	`m4_fm_only`	**0.0667**	201.4	0.897	**0.314**
	`m5_fm_rules`	**0.0667**	201.6	0.897	**0.314**
S2 (cross-protocol)	`m1_persistence`	**0.0150**	**27.3**	0.000	0.521
	`m3_evolvegcn`	0.0116	196.4	0.000	0.189
	`m4_fm_only`	0.0151	50.3	0.000	**0.525**
	`m5_fm_rules`	0.0151	52.1	0.000	**0.525**
S3 (bridge)	`m1_persistence`	**0.0079**	0.0	0.000	0.445
	`m3_evolvegcn`	0.0156	0.0	0.000	0.269
	`m4_fm_only`	0.0080	0.0	0.000	0.445
	`m5_fm_rules`	0.0080	0.0	0.000	0.445
S4 (stablecoin)	`m1_persistence`	0.0100	0.0	0.034	0.536
	`m3_evolvegcn`	0.0200	0.0	0.034	0.293
	`m4_fm_only`	**0.0099**	0.0	0.034	0.536
	`m5_fm_rules`	**0.0099**	0.0	0.034	0.536
S5 (lending)	`m1_persistence`	0.0100	0.0	0.000	**0.602**
	`m3_evolvegcn`	0.0538	0.0	1.000	0.330
	`m4_fm_only`	**0.0099**	0.0	0.000	**0.602**
	`m5_fm_rules`	**0.0099**	0.0	0.000	**0.602**

Table 6: `b6_robustness` per-regime detail. All metrics at h=4 on the 2025 test split under five degradation regimes. PageRank, HHI, and Gini MAE in 10^{-3} units. Relative degradation is signed against the clean `b1_forecast` baseline; negative values mean the predictor is no worse than on clean data. Best per metric (lowest MAE, highest rank correlation, smallest $|\Delta_{\mathrm{rel}}|$) in **bold**. Persistence ignores node features, so its `noisy_features_01` and `missing_features_20` rows degenerate to the clean numbers (and $\Delta_{\mathrm{rel}} = 0$).

Regime	Method	PR MAE (↓)	HHI MAE (↓)	Gini MAE (↓)	Rank Corr (↑)	Δ_{rel} (~0)
low_data_10pct	`m1_persistence`	15.94	0.0591	**0.0023**	**0.552**	$+0.766$
	`m3_evolvegcn`	15.23	0.0999	0.1301	0.535	$+0.276$
	`m4_fm_only`	12.83	0.0586	0.0027	0.528	$+0.612$
	`m5_fm_rules`	**12.71**	**0.0587**	0.0027	0.528	$\mathbf{+0.607}$
low_data_25pct	`m1_persistence`	**9.46**	0.0282	**0.0020**	**0.577**	-0.093
	`m3_evolvegcn`	10.22	0.0486	0.1281	0.558	$\mathbf{-0.027}$
	`m4_fm_only`	9.86	**0.0279**	0.0024	0.553	-0.126
	`m5_fm_rules`	9.95	**0.0279**	0.0024	0.553	-0.126
partial_graph_30	`m1_persistence`	**9.63**	0.0350	**0.0019**	**0.554**	$+0.068$
	`m3_evolvegcn`	10.36	0.0488	0.1236	0.537	$\mathbf{-0.048}$
	`m4_fm_only`	12.52	**0.0350**	0.0022	0.500	$+0.082$
	`m5_fm_rules`	12.45	**0.0350**	0.0022	0.499	$+0.077$
noisy_features_01	`m1_persistence`	**9.77**	**0.0322**	**0.0018**	**0.584**	$\mathbf{0.000}$
	`m3_evolvegcn`	10.68	0.0489	0.1326	0.563	$\mathbf{0.000}$
	`m4_fm_only`	11.38	0.0323	0.0021	0.554	-0.005
	`m5_fm_rules`	11.35	0.0323	0.0021	0.554	-0.007
missing_features_20	`m1_persistence`	**9.77**	**0.0322**	**0.0018**	**0.584**	$\mathbf{0.000}$
	`m3_evolvegcn`	10.68	0.0489	0.1326	0.563	$\mathbf{0.000}$
	`m4_fm_only`	12.11	0.0323	0.0023	0.559	$+0.017$
	`m5_fm_rules`	12.08	0.0323	0.0023	0.559	$+0.014$

Table 7: `b2_warning` per-event detail for the shared weighted-degree monitor (`h1_weighted_degree`) across alert budgets $K \in \{5, 10, 20\}$. Recall and F1-warning shown in 10^{-3} units (the full DeFi corpus contains ∼4,300 protocols, so per-week recall denominators are large). Lead time is event-level and not budget-sensitive: 5 weeks for Terra/Luna and FTX, 4 weeks for SVB-USDC. `b2_warning` sits upstream of the predictor choice, so a single monitor is shared across all methods (see §5).

Event	K	Prec. (↑)	Rec. $\times 10^3$ (↑)	$\mathrm{F1_w}$ $\times 10^3$ (↑)	Stab. (↑)	$\lvert\mathcal{A}\rvert$
Terra/Luna	5	0.60	1.81	3.61	0.44	11
	10	0.70	3.26	6.49	0.62	19
	20	0.65	7.97	15.74	0.74	36
FTX	5	0.60	1.33	2.65	0.64	10
	10	0.70	2.39	4.77	0.72	18
	20	0.55	4.79	9.49	0.72	39
SVB-USDC	5	1.00	1.81	3.62	0.55	10
	10	1.00	2.82	5.62	0.75	15
	20	1.00	5.64	11.22	0.81	30

Table 8: Per-regime data-health gate ablation isolating A1 from A2. Each row holds $\tau_{\text{conf}}=0$ and sweeps τ_{data}. "*strict*" = $\tau_{\text{data}}=0.85$, "*on*" = 0.70, "*off*" = 0.00. The four regimes apply targeted edge or feature degradation on the clean 2025 test split (`n_weeks` = 29): `extreme` masks both topology and features at 90%, `topo` removes 90% of edges only, `feat` corrupts 90% of node features only, `severe` masks both at 80%. Numbers expand Panel B of Table 9: under degradation the gate-off setting produces FIR up to 0.541, while *on*/*strict* can suppress intervention tickets via safe-mode and drive FIR to 0.

Regime	Gate	τ_{data}	$\overline{\text{dh}}$	Safe-mode (%)	Ticket Prec. (↑)	FIR (↓)	Interventions
extreme	strict	0.85	0.587	100	0.510	**0.000**	0
	on	0.70	0.587	100	0.545	**0.000**	0
	off	0.00	0.587	0	0.490	0.510	57
extreme_topo	strict	0.85	0.637	100	0.517	**0.000**	0
	on	0.70	0.637	100	0.462	**0.000**	0
	off	0.00	0.637	0	0.559	0.441	58
extreme_feat	strict	0.85	0.755	100	0.503	**0.000**	0
	on	0.70	0.755	0	0.510	0.267	5
	off	0.00	0.755	0	0.503	0.267	5
severe	strict	0.85	0.751	100	0.490	**0.000**	0
	on	0.70	0.751	0	0.538	0.400	25
	off	0.00	0.751	0	0.538	0.541	24

Table 9: Component-level ablation of Claw-System on `b5_decision`. Panel A shows clean 2025 test-split ablations against the `m5_fm_rules` baseline; Panels B–C report targeted stress runs for mechanisms that are dormant on clean data. A2 (confidence gating) and A3 (scenario engine) are load-bearing in the clean setting, while A1 (data-health gating) and A6 (multi-horizon monitoring) are safety-reserve mechanisms whose effects appear under data degradation and crisis-window dynamics, respectively.

Panel	Ablation	Configuration	Ticket Prec.	FIR	Notes
A: clean test	—	full `m5_fm_rules`	0.600	0.000	baseline
	A1	disable data-health gate (τ_{data}=0)	0.600	0.000	no effect (clean data)
	A2	disable confidence gate (τ_{conf}=0)	0.600	0.429	FIR shoots up
	A3	skip scenario engine	0.000	0.000	target extraction collapses
	A6	single-horizon forecasting (h=4)	0.600	0.000	no effect (clean data)
B: degraded data	A1 off	80–98% feature/edge mask, τ_{conf}=0	—	0.27–0.60	false interventions emerge
	A1 on	same degradation, safe-mode triggered	—	0.000	intervention suppression
C: crisis period	A6: single	horizons = $\{4\}$, crisis windows	0.533–0.624	0.000	alerts: 19–23
	A6: multi	horizons = $\{1, 4, 8, 12\}$, same windows	0.533–0.624	0.000	alerts: 73–98 ($\approx 4\times$)